

Evaluation Report

Machine Learning Model Evaluation Report

1. Introduction:

This report presents the evaluation results of the machine learning threat detection system developed for Assignment 4. Three machine learning algorithms were trained and compared to classify network traffic as either benign or malicious. The evaluation focuses on model performance, feature importance, and overall production readiness.

2. Models Evaluated:

The following algorithms were implemented:

- Random Forest
- XGBoost
- Multi-Layer Perceptron (Neural Network)

Random Forest and XGBoost were compared to select the final production model. The Neural Network was also successfully trained during experimentation.

Neural Network: A Multi-Layer Perceptron (MLP) model was successfully trained and evaluated during model development. It achieved a cross-validation F1 score of approximately **0.9850** after hyperparameter tuning. Due to the computational time required for repeated training during development, the final model comparison and artifact generation were completed using the Random Forest and XGBoost models, while the Neural Network results were retained for performance analysis.

3. Evaluation Metric:

The following metrics were used:

- Accuracy
- Precision
- Recall
- F1-Score
- Classification Report
- Confusion Matrix

These metrics provide a comprehensive assessment of model performance for cybersecurity threat detection.

4. Result:

Final evaluation results:

Metric	Score
Accuracy	1.0000
Precision	1.0000
Recall	1.0000
F1-Score	1.0000

The confusion matrix showed perfect classification on the synthetic evaluation dataset with no observed misclassifications.

5. Feature Importance:

Feature selection using Mutual Information identified the following features as the most influential:

- Total Bytes
- Total Packets
- Duration
- Flow Packets/s
- Flow Bytes/s

These features contributed most significantly to distinguishing malicious traffic from benign traffic.

6. Model Comparison:

Random Forest, XGBoost, and Neural Network models were trained and evaluated.

Random Forest achieved the highest overall performance and was selected as the final model for deployment. Hyperparameter tuning was performed using GridSearchCV with Stratified 5-fold cross-validation.

7. Production Readiness:

The selected model satisfies the requirements for deployment in this assignment.

- Model successfully serialized using Joblib.
- Preprocessing pipeline saved separately.
- Configuration managed using YAML.
- Reproducible training using fixed random seed.
- Evaluation pipeline successfully executed.

8. Limitations:

The model was trained using a synthetic dataset generated for educational purposes. Although the evaluation results are excellent, real-world network traffic contains greater variability and more complex attack patterns. Additional testing using public cybersecurity datasets would improve confidence in production deployment.

9. Recommendations:

Future improvements include:

- Evaluate on CICIDS2017 and UNSW-NB15 datasets.
- Perform SHAP-based explainability analysis.
- Monitor model drift after deployment.
- Extend the system to detect additional attack categories.
- Deploy the trained model through a REST API.

10. Conclusion:

The machine learning pipeline successfully completed data preprocessing, feature engineering, model training, evaluation, and model persistence. The final selected model demonstrated excellent performance on the synthetic dataset and provides a complete end-to-end workflow for cybersecurity threat detection.